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Chapter 11: Reproduction

Reproduction: The ability of an organism to produce new offspring of its own type. It is a unique characteristic of life: not essential for survival of the individual but required for survival of the species.

- Human reproduction needs **internal fertilization**. The reproductive system is unique in two ways: (1) it is not functional until **puberty**, turned on by sex hormones (other systems work at birth); (2) it is quite different in male and female (other systems differ slightly).

11.1 Male reproductive system (SLO 1)

- Main function: to **produce and maintain sperms**. Parts: **gonads** (testes), **accessory ducts** (epididymis, ductus deferens, ejaculatory duct, urethra), **accessory glands** (seminal vesicles, prostate gland, bulbourethral glands) and **copulatory organ** (penis).

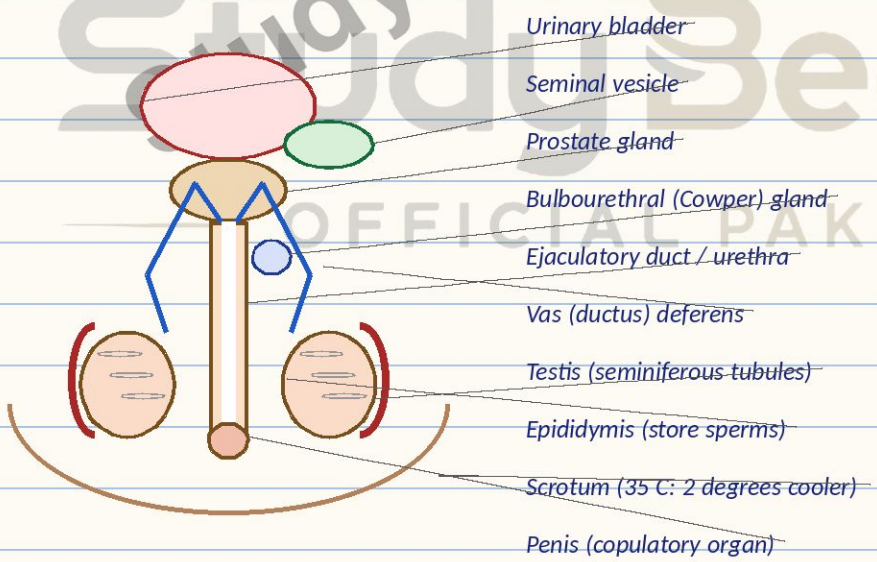


Fig: Human male reproductive system (lateral view)

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Structure	Description and function
Testes (gonads)	Outside the abdomen in a skin pouch, the **scrotum**; each testis has 250-300 lobules, each with 1-4 tightly coiled **seminiferous tubules** where sperms are made. Temperature of scrotum is about 35 C, two degrees cooler than the body: sperm production (spermatogenesis) is most efficient there.
Rete testis and efferent ductules	Sperms move through seminiferous tubules into the rete testis for further maturation and leave the testis by efferent ductules.
Epididymis	Coiled on the outer surface of the testis; **transport and storage** of sperms (stored before ejaculation).
Vas (ductus) deferens	Continues from epididymis; joins the duct of the seminal vesicle to form the short **ejaculatory duct**, which enters the prostate and opens into the urethra.
Urethra	Also called **urinogenital duct**: carries both urine and semen.
Seminal vesicles (pair)	At the junction of sperm duct and ejaculatory duct; secrete nutritive fluid of semen.
Prostate gland	Encircles the urethra just below the bladder.
Bulbourethral (Cowper's) glands (pair)	At the junction of ejaculatory duct and urethra.
Penis	Copulatory organ; tissue fills with blood to cause an **erection**.

- **Semen:** white, sticky mixture of sperms and secretions of accessory glands; the liquid gives nutrients, protection and a transport medium. About **2-5 ml** per ejaculation with **20 to 150 million sperms per ml**.

Spermatogenesis (extra reading)

- Formation of sperms in the **germinal epithelium** (inner epithelium) of seminiferous tubules. A healthy adult male makes about **400 million sperms per day**.

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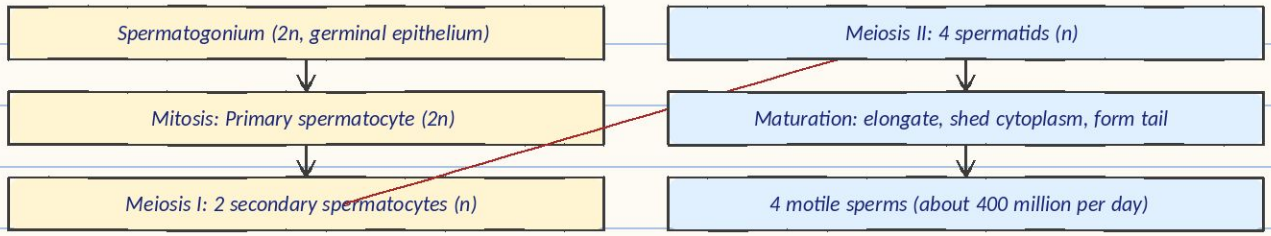


Fig: Spermatogenesis (in seminiferous tubules)

- *Spermatogonia divide by mitosis to primary spermatocytes; each primary spermatocyte completes meiosis I to two haploid **secondary spermatocytes**; meiosis II gives **spermatids** (round, non-motile, haploid). Spermatid elongates, sheds excess cytoplasm and forms a tail = motile **sperm**.*

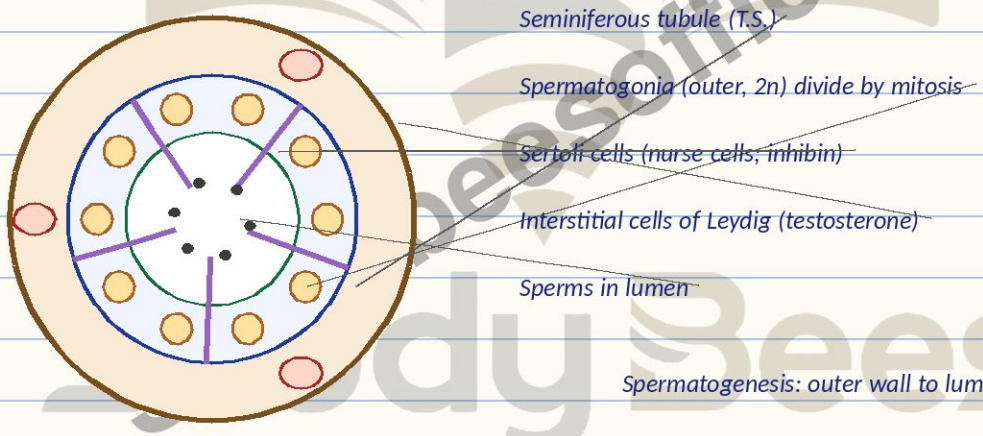


Fig: Section of seminiferous tubule

Male reproductive hormones (SLO 2)

- ***Hormones** are chemical messengers made by glands, carried by blood to act on other organs, for growth, reproduction and well-being. Male and female cycles are controlled by hormones of the **hypothalamus** and **anterior pituitary**.*

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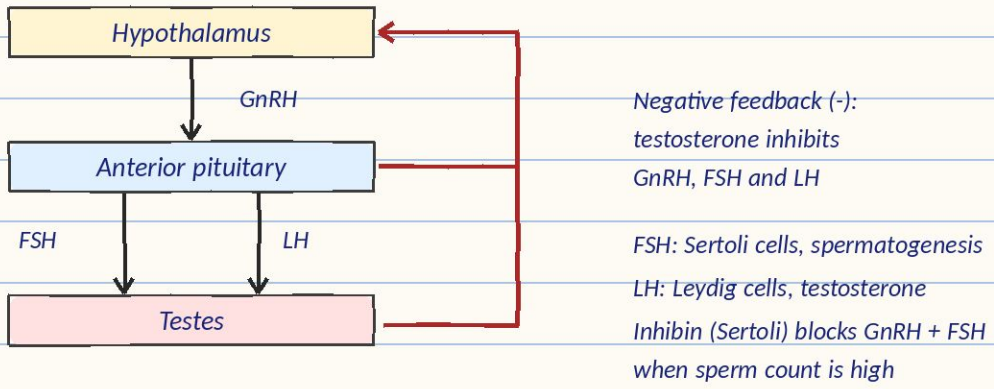


Fig: Hormonal control of male reproduction

Hormone	**Source and function**
GnRH (gonadotropin-releasing hormone)	Hypothalamus; at puberty it makes the anterior pituitary release FSH and LH.
FSH (follicle stimulating hormone)	Anterior pituitary; enters testes and stimulates **Sertoli cells** to begin spermatogenesis.
LH (luteinizing hormone)	Anterior pituitary; stimulates **interstitial cells of Leydig** to make and release testosterone.
Testosterone	Leydig cells; stimulates spermatogenesis and causes **secondary sexual characters**: deepening of voice, growth of facial, axillary and pubic hair etc.
Inhibin	Sertoli cells; released when sperm count is too high; inhibits GnRH and FSH so spermatogenesis slows. If sperm count falls to 20 million/ml, inhibin stops and the count rises.

- ***Negative feedback:*** *rising testosterone acts on hypothalamus and anterior pituitary to inhibit release of GnRH, FSH and LH. FSH and LH are produced in* ***both sexes*** *though named after female functions.*

11.1.3 Female reproductive system (SLO 3)

- *The female role is far more complex: she produces gametes and must nurture the embryo for about* ***nine months****. Parts: a pair of* ***ovaries****,* ***oviducts****,* ***uterus***

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*and **vagina**.*

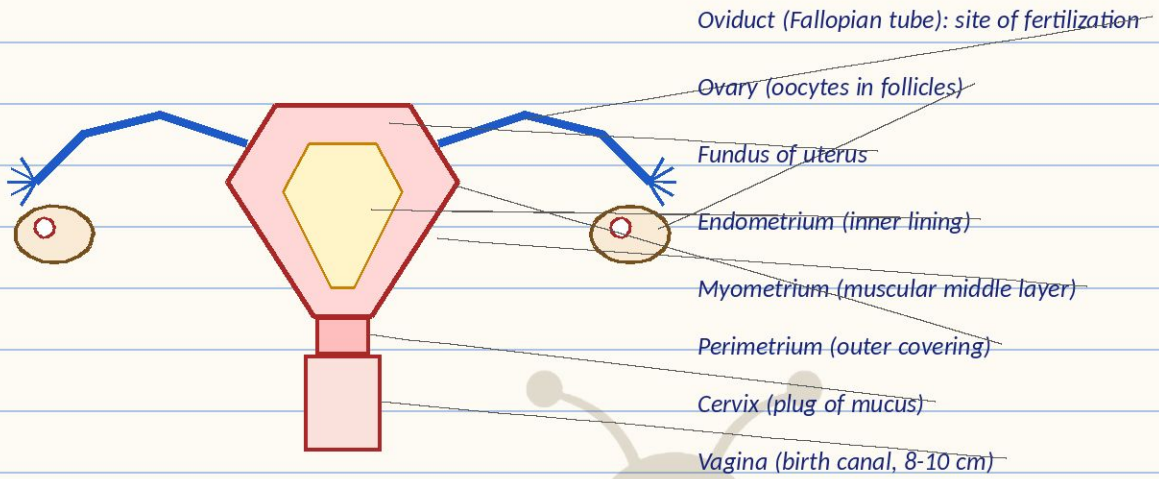


Fig: Human female reproductive system (frontal view)

Structure	***Description and function***
Ovaries	*Female gonads; solid, ovoid, 3-5 cm long and 2-3 cm wide, held by ligaments in the peritoneal cavity; produce ova and hormones. Contain many saclike **ovarian follicles**, each with an immature egg (**oocyte**). Monthly one ripening follicle ejects its oocyte = **ovulation**; the ruptured follicle becomes the glandular **corpus luteum**.*
Oviducts (Fallopian tubes)	*About 10 cm long; receive the ovulated oocyte; **site where fertilization generally occurs**; carry the developing ovum toward the uterus.*
Uterus (womb)	*Hollow muscular inverted-pear-shaped organ with **fundus, body and cervix**. Wall: **perimetrium** (outer thin covering), **myometrium** (middle thick smooth muscle; contracts at childbirth), **endometrium** (inner spongy lining where the embryo implants). Function: receive, retain and nourish the fertilized ovum.*
Cervix	*Narrow entrance from vagina to the uterus; normally blocked by a plug of mucus.*

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Structure	Description and function
Vagina	Thin walled tube, 8-10 cm, from cervix to the body exterior; **birth canal** and passage for menstrual flow; urethra is embedded in its anterior wall.

- ***Functions of the female system:*** *production of egg (**oogenesis**), fertilization, development of embryo, and production of female sex hormones.*

Oogenesis (extra reading)

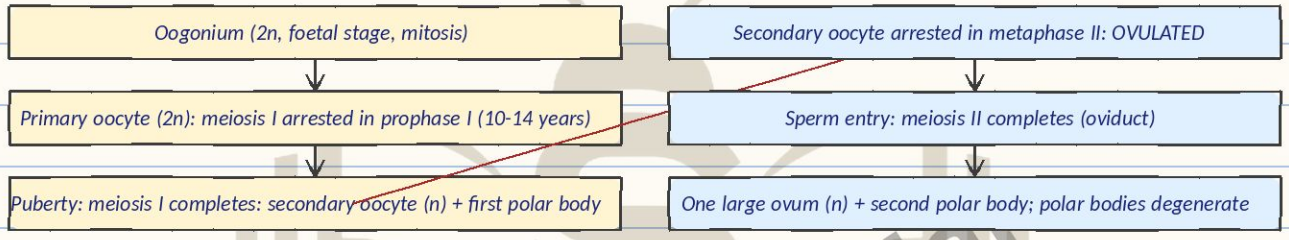


Fig: Oogenesis (unequal divisions keep nutrients in the ovum)

- *Starts in the foetus: **oogonia** (diploid stem cells) multiply by mitosis, grow and become **primary oocytes** surrounded by follicle cells. Primary oocytes begin meiosis I but arrest in prophase I for at least 10-14 years. At puberty a few are recruited each month and only one continues: it forms two dissimilar haploid cells: the large **secondary oocyte** (nearly all cytoplasm) and small **first polar body**.*
- *The secondary oocyte arrests in metaphase II and is ovulated. Without sperm it deteriorates; if a sperm penetrates it quickly completes meiosis II in the oviduct to give one large **ovum** and a tiny **second polar body**. Unequal division ensures the fertilized egg has ample nutrients for its 6-7 day journey to the uterus; polar bodies degenerate.*

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11.2 Female reproductive (menstrual) cycle and hormonal regulation (SLO 4)

- *The sequence of all reproductive events in the female = **female reproductive (menstrual) cycle**, divided into the **ovarian cycle** (events in ovary) and **uterine cycle** (events in uterus), coordinated by pituitary **gonadotropins**. Three phases: menstrual, proliferative and secretory.*

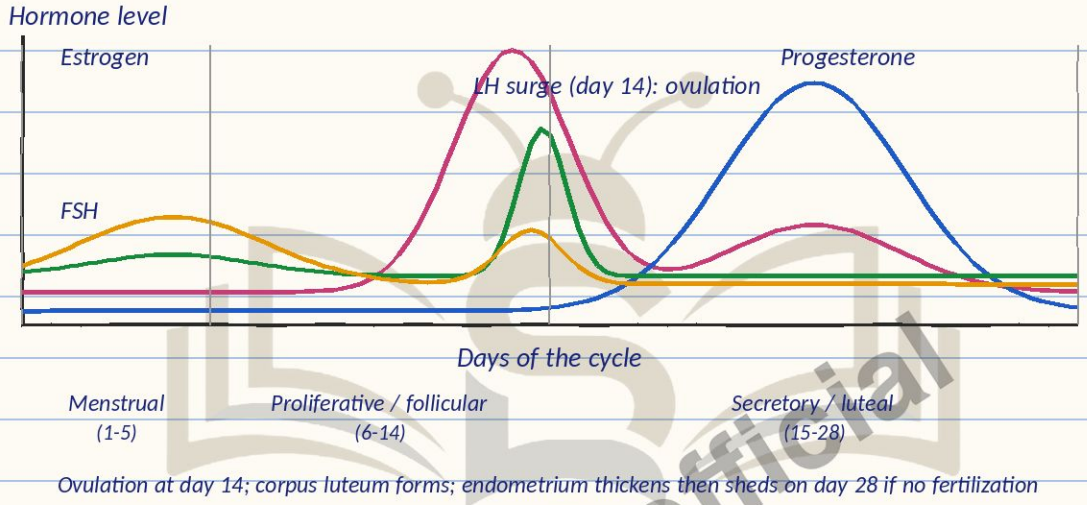


Fig: Human female reproductive cycle and hormones

Phase (days)	Events
Menstrual (days 1-5)	Uterus sheds all but the deepest part of the endometrium: the thick, hormone-dependent functional layer detaches; bleeding for 3-5 days (menstrual flow). Ovarian hormones are at lowest levels; gonadotropins start rising; FSH begins to rise.
Proliferative / pre-ovulatory / follicular (days 6-14)	FSH stimulates a few follicles; they compete; one dominant follicle (**Graafian follicle**) matures, others disintegrate (**follicle atresia**). FSH makes the Graafian follicle secrete **estrogen**, which thickens and vascularizes the endometrium and thins the cervical mucus so sperms can pass. Estrogen has negative feedback on FSH (FSH falls) which signals the anterior pituitary to release **LH** at day 14: **ovulation** (less than 5 minutes). LH converts the ruptured follicle to the yellowish **corpus luteum**.

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Phase (days)	Events
Secretory / post-ovulatory / luteal (days 15-28)	Corpus luteum secretes rising **progesterone** acting on the estrogen-primed endometrium: arteries elaborate, uterine glands enlarge, coil and secrete nutritious **glycogen** to sustain the embryo until implantation. If fertilization does not occur, the corpus luteum degenerates as LH falls; progesterone falls; endometrial cells die; menstruation starts on day 28.

- ***Menarche** = first menstruation; **menopause** = cessation of the cycle around **50 years**. Cyclic menstruation is an indicator of the normal reproductive life of a female (between menarche and menopause).*

Infertility and IVF (exercise topic; standard notes)

***Infertility:** Inability to conceive after a year of regular unprotected intercourse.*

- ***Male causes:** low sperm count or poor sperm motility / abnormal shape; blocked ducts (epididymis, vas deferens); low testosterone; infections; varicocele; high scrotal temperature; smoking, alcohol, drugs; stress.*
- ***Female causes:** failure to ovulate (hormonal imbalance, PCOS); blocked or damaged oviducts; uterine problems (fibroids, endometriosis); cervical mucus problems; infections; age; stress and lifestyle.*

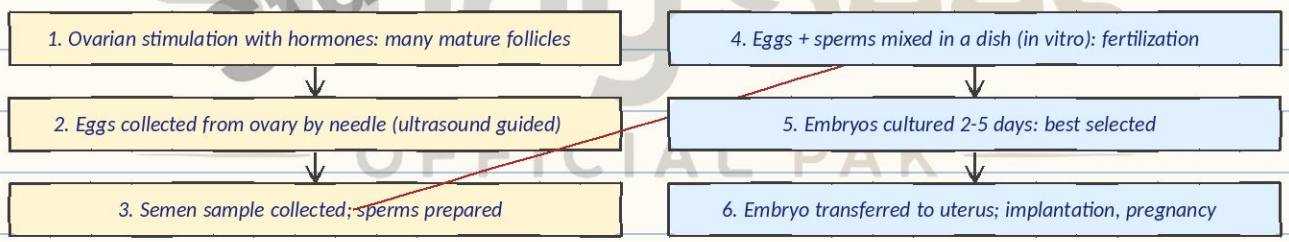


Fig: In vitro fertilization (IVF) - test tube baby technique

- ***IVF:** fertilization outside the body (in vitro = in glass); the baby is called a **test tube baby**. Used when oviducts are blocked or sperm count is low.*

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Chapter 11: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define reproduction. Why is it important?

Ans: Ability of an organism to produce offspring of its own type; it is not necessary for the survival of the individual but is required for survival of the species.

Q2. [SLO 1] In which two ways is the human reproductive system unique?

Ans: It does not become functional until puberty (turned on by sex hormones), and it is very different in male and female.

Q3. [SLO 1] List the structures of the male reproductive system.

Ans: Gonads (testes), accessory ducts (epididymis, ductus deferens, ejaculatory duct, urethra), accessory glands (seminal vesicles, prostate, bulbourethral glands) and penis.

Q4. [SLO 1] Why are the testes located outside the body?

Ans: Sperm production is most efficient at about 35 C, two degrees cooler than body temperature.

Q5. [SLO 1] What are seminiferous tubules?

Ans: Tightly coiled tubules (1-4 per lobule; 250-300 lobules per testis) in which sperms are produced.

Q6. [SLO 1] What is the function of the epididymis?

Ans: Transport and storage of sperms before ejaculation.

Q7. [SLO 1] Why is the male urethra called the urinogenital duct?

Ans: It carries both urine and semen.

Q8. [SLO 1] List the structures through which a sperm passes from seminiferous tubules to the fallopian tube.

Ans: Seminiferous tubules, rete testis, efferent ductules, epididymis, vas deferens, ejaculatory duct, urethra, vagina, cervix, uterus, fallopian tube.

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Q9. [SLO 1] Name the accessory glands of the male system.

Ans: Seminal vesicles, prostate gland and bulbourethral (Cowper's) glands.

Q10. [SLO 1] What is semen? How much is released and how many sperms does it have?

Ans: White sticky mixture of sperms and accessory gland secretions; about 2-5 ml with 20 to 150 million sperms per ml.

Q11. [SLO 1] What is the function of the liquid part of semen?

Ans: It provides nutrients and protection to sperms and acts as their transport medium.

Q12. [SLO 1] Define spermatogenesis. Where does it take place?

Ans: Formation of sperms; in the germinal epithelium of seminiferous tubules.

Q13. [SLO 1] Differentiate primary and secondary spermatocytes.

Ans: Primary spermatocyte: diploid, formed by mitosis, undergoes meiosis I. Secondary spermatocyte: haploid, formed after meiosis I, undergoes meiosis II.

Q14. [SLO 1] Differentiate spermatids and sperms.

Ans: Spermatids are round, non-motile haploid cells; they mature (shed cytoplasm, form tail) into motile sperms.

Q15. [SLO 1] How many sperms does a man make daily?

Ans: About 400 million sperms per day.

Q16. [SLO 1] Why are so many sperms produced in the male and so few ova in the female?

Ans: Sperms are tiny and many are lost in the female tract; large numbers are needed to reach and fertilize the egg; the female makes one large nutrient-rich ovum a month for the embryo.

Q17. [SLO 1] Why must many sperms be produced when only one is needed for fertilization?

Ans: Many sperms die or are lost on the way; many are needed to digest the egg coverings and to reach the oviduct.

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Q18. [SLO 1] Enlist reasons of human male infertility.

Ans: Low sperm count, poor motility or abnormal sperms, blocked ducts, low testosterone, infections, high scrotal temperature, smoking, alcohol and stress.

Q19. [SLO 1] Differentiate sertoli cell and Leydig cell.

Ans: Sertoli cells (in seminiferous tubules) nourish sperms and make inhibin; Leydig cells (interstitial) make testosterone.

Q20. [SLO 2] What are hormones?

Ans: Chemical messengers made by glands and carried by blood to act on other organs.

Q21. [SLO 2] What does GnRH do?

Ans: Made by the hypothalamus; makes the anterior pituitary release FSH and LH.

Q22. [SLO 2] What are the actions of FSH and LH in males?

Ans: FSH stimulates Sertoli cells to start spermatogenesis; LH stimulates Leydig cells to make testosterone.

Q23. [SLO 2] What is the role of testosterone?

Ans: Stimulates spermatogenesis and causes secondary sex characters such as deep voice and facial, axillary and pubic hair.

Q24. [SLO 2] Write the role of inhibin.

Ans: Released by Sertoli cells when the sperm count is high; inhibits GnRH and FSH so spermatogenesis slows.

Q25. [SLO 2] Explain negative feedback in the male reproductive system.

Ans: Rising testosterone acts on the hypothalamus and anterior pituitary to inhibit GnRH, FSH and LH release, keeping hormone levels steady.

Q26. [SLO 2] Why are FSH and LH not female hormones only?

Ans: They are named after female functions but are produced in both sexes and control reproduction in both.

Q27. [SLO 3] List the structures of the female reproductive system.

Ans: A pair of ovaries, oviducts (Fallopian tubes), uterus (with cervix) and vagina.

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Q28. [SLO 3] What are ovarian follicles?

Ans: Tiny saclike structures in the ovary, each containing an immature egg called an oocyte.

Q29. [SLO 3] Define ovulation and corpus luteum.

Ans: Ovulation: release of the oocyte from a ripe follicle. Corpus luteum: the glandular structure formed from the ruptured follicle.

Q30. [SLO 3] What is the site of fertilization?

Ans: The oviduct (Fallopian tube).

Q31. [SLO 3] Describe the three layers of the uterine wall.

Ans: Perimetrium (outer thin covering), myometrium (middle thick muscle) and endometrium (inner spongy lining where embryo implants).

Q32. [SLO 3] Differentiate perimetrium and endometrium.

Ans: Perimetrium is the outermost thin layer; endometrium is the inner spongy lining that receives the embryo and sheds in menstruation.

Q33. [SLO 3] What is the function of the myometrium?

Ans: It contracts rhythmically during childbirth to expel the baby.

Q34. [SLO 3] What is the cervix?

Ans: The narrow entrance from the vagina to the uterus; normally blocked by a plug of mucus.

Q35. [SLO 3] Why is the vagina called the birth canal?

Ans: It is the passageway for delivery of the infant and for menstrual flow.

Q36. [SLO 3] Write the functions of the female reproductive system.

Ans: Production of egg (oogenesis), fertilization, development of embryo and production of female sex hormones.

Q37. [SLO 3] Define oogenesis.

Ans: The process of egg formation in females; it begins in the foetus.

Q38. [SLO 3] Differentiate primary and secondary oocyte.

Ans: Primary oocyte: diploid, arrested in prophase I. Secondary oocyte: haploid large cell formed after meiosis I (with first polar body), arrested in metaphase II and ovulated.

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Q39. [SLO 3] Why does oogenesis have unequal division of cytoplasm?

Ans: So that the fertilized egg has ample nutrients for its 6-7 day journey to the uterus; polar bodies degenerate.

Q40. [SLO 3] What are polar bodies?

Ans: Tiny cells formed with almost no cytoplasm during meiosis of the oocyte; they degenerate.

Q41. [SLO 3] Differentiate spermatogenesis and oogenesis.

Ans: Spermatogenesis gives four motile sperms from one cell, from puberty continuously; oogenesis gives one ovum and polar bodies, starts in foetus and is arrested for years.

Q42. [SLO 3] How does a zygote differ from an ovum?

Ans: A zygote is diploid (formed by fertilization); an ovum is haploid.

Q43. [SLO 4] Define menstrual cycle. Name its phases and days.

Ans: Sequence of reproductive events in females: menstrual (days 1-5), proliferative (6-14) and secretory (15-28) phases.

Q44. [SLO 4] What is the ovarian cycle and the uterine cycle?

Ans: Ovarian cycle: events in the ovary. Uterine cycle: events in the uterus; coordinated by gonadotropins.

Q45. [SLO 4] What happens during menstruation?

Ans: The uterus sheds the functional layer of the endometrium with blood for 3-5 days.

Q46. [SLO 4] What is a Graafian follicle?

Ans: The one dominant mature follicle in which oogenesis occurs and which secretes estrogen.

Q47. [SLO 4] What is follicle atresia?

Ans: Disintegration of the non-dominant follicles that stop growing.

Q48. [SLO 4] What does estrogen do in the proliferative phase?

Ans: Thickens and vascularizes the endometrium, thins cervical mucus for sperm passage, and has negative feedback on FSH.

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Q49. [SLO 4] What triggers ovulation?

Ans: The sudden surge of LH from the anterior pituitary at about day 14 when estrogen is high.

Q50. [SLO 4] What is the role of the corpus luteum in the menstrual cycle?

Ans: It secretes progesterone which prepares the endometrium (glandular, nutritious) for implantation; it degenerates if there is no fertilization.

Q51. [SLO 4] Why does menstruation begin on day 28?

Ans: The corpus luteum degenerates as LH falls, progesterone falls, and the unsupported endometrium dies and sheds.

Q52. [SLO 4] What are the actions of GnRH, FSH and LH in female reproduction?

Ans: GnRH releases FSH and LH; FSH grows follicles and estrogen; LH causes ovulation and forms the corpus luteum.

Q53. [SLO 4] Define menarche and menopause.

Ans: Menarche: first menstruation. Menopause: cessation of the cycle around 50 years of age.

Q54. [SLO 4] What changes occur in ovulation and menstruation during pregnancy?

Ans: Ovulation and menstruation stop because the corpus luteum persists and hormones keep the endometrium for the embryo (standard).

Q55. [SLO 4] What is IVF?

Ans: In vitro fertilization: eggs and sperms are mixed in a dish outside the body and the embryo is transferred to the uterus; gives test tube babies.

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Chapter 11: Long Questions with Answers (SLO based)

Q1. [SLO 1] Describe the structure of the human male reproductive system. Identify the function of each part.

Ans:

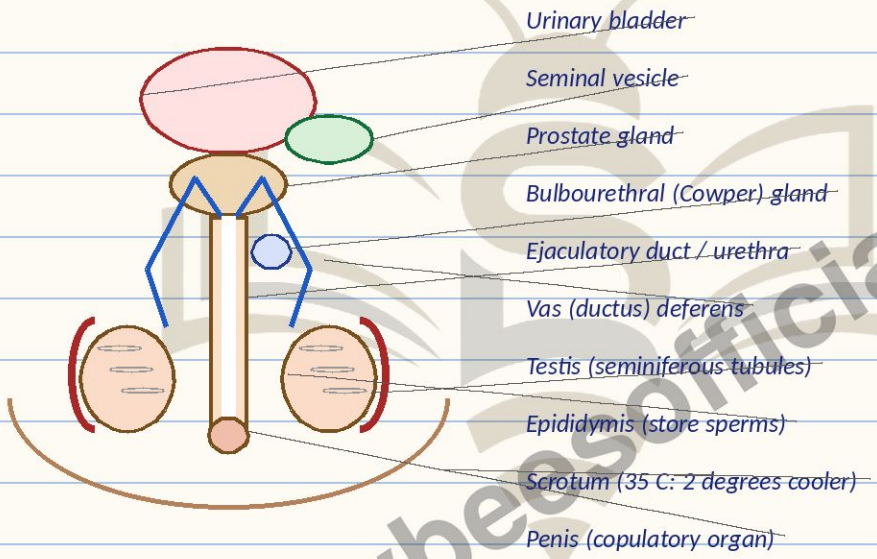


Fig: Human male reproductive system (lateral view)

Part	Function
Testes	Sperm production (spermatogenesis) and testosterone
Scrotum	Keeps testes about 2 degrees cooler (35 C)
Epididymis	Transport and storage of sperms
Vas deferens / ejaculatory duct	Carry sperms to the urethra
Urethra	Urine and semen passage
Seminal vesicles, prostate, Cowper's glands	Secrete the liquid part of semen
Penis	Copulatory organ (erection by filling of tissues with blood)

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Q2. [SLO 1] Describe spermatogenesis in humans.

Ans:

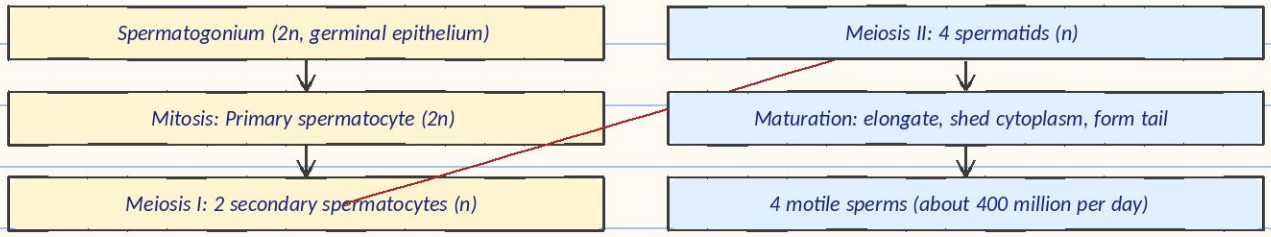


Fig: Spermatogenesis (in seminiferous tubules)

- *In germinal epithelium of seminiferous tubules: spermatogonium, mitosis, primary spermatocyte (2n), meiosis I, two secondary spermatocytes (n), meiosis II, four spermatids, maturation (tail formed, cytoplasm shed) to sperms; about 400 million per day.*

Q3. [SLO 2] Describe the principal reproductive hormones of the human male and explain their role.

Ans:

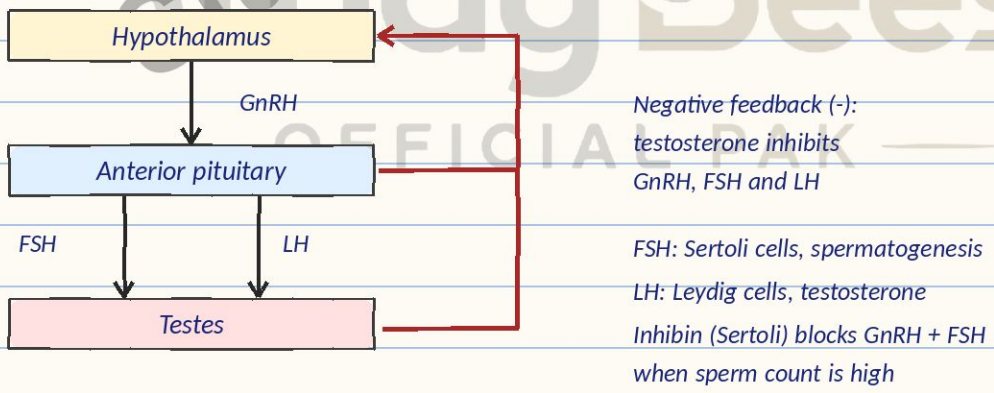


Fig: Hormonal control of male reproduction

- *GnRH (hypothalamus) leads to FSH and LH (anterior pituitary). FSH: Sertoli cells, spermatogenesis. LH: Leydig cells, testosterone. Testosterone:*

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spermatogenesis and secondary sex characters (voice, facial, axillary and pubic hair).

- *Negative feedback: testosterone inhibits GnRH, FSH and LH; inhibin (Sertoli) inhibits GnRH and FSH when sperm count is high; below 20 million/ml inhibin stops.*

Q4. [SLO 3] Explain the structure of the human female reproductive system. Identify the function of each part.

Ans:

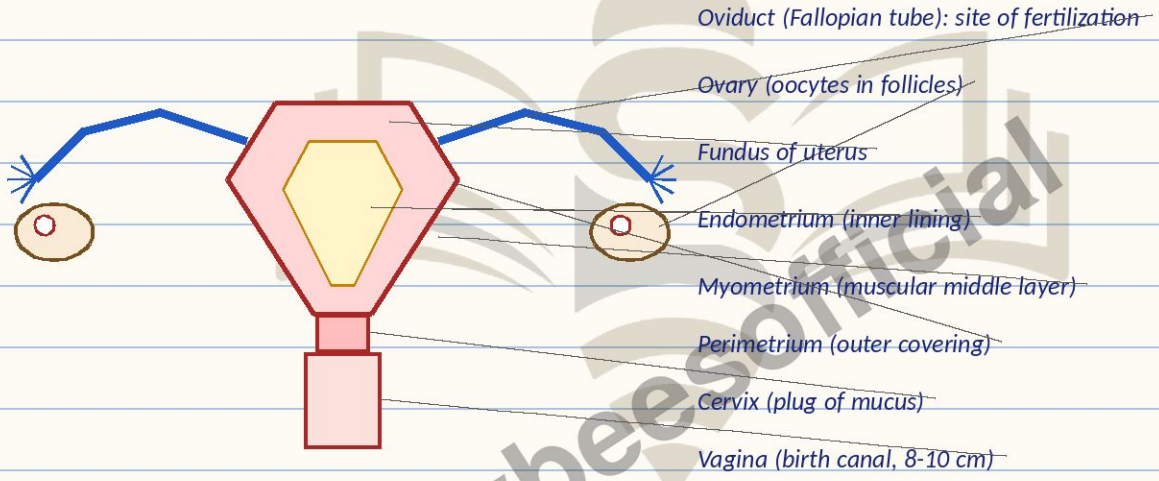


Fig: Human female reproductive system (frontal view)

Part	***Function***
Ovaries	*Produce ova (oocytes) and hormones; ovulation*
Oviducts	*Receive oocyte; site of fertilization; carry ovum to uterus*
Uterus	*Receive, retain and nourish fertilized ovum; perimetrium, myometrium (childbirth contractions), endometrium (implantation)*
Cervix	*Entrance to uterus, mucus plug*
Vagina	*Birth canal; menstrual flow passage*

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Q5. [SLO 3] Describe oogenesis in humans.

Ans:

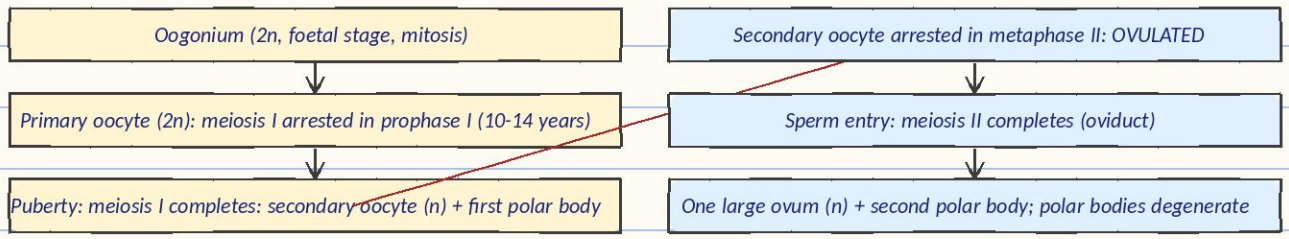


Fig: Oogenesis (unequal divisions keep nutrients in the ovum)

- *Oogonium to primary oocyte (arrested prophase I for 10-14 years); at puberty one completes meiosis I each month: secondary oocyte + first polar body; secondary oocyte arrests in metaphase II and is ovulated; sperm entry completes meiosis II: ovum + second polar body. Unequal division stores nutrients in the ovum.*

Q6. [SLO 4] Describe the events of a menstrual cycle and explain its hormonal regulation.

Ans:

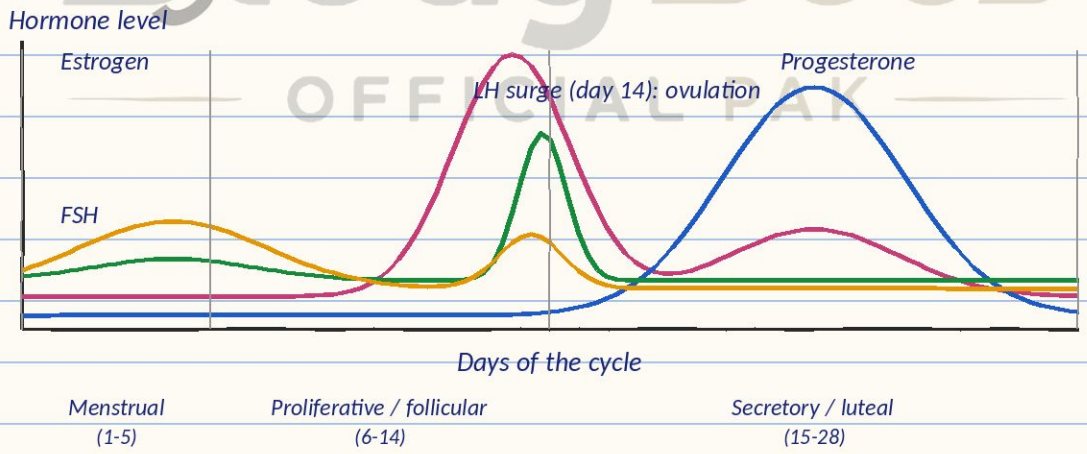


Fig: Human female reproductive cycle and hormones

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- ***Menstrual (1-5):*** *endometrium shed with bleeding (3-5 days); FSH begins to rise.*
- ***Proliferative (6-14):*** *FSH grows follicles, Graafian follicle makes estrogen; endometrium thickens; estrogen inhibits FSH; LH surge on day 14 causes ovulation; corpus luteum forms.*
- ***Secretory (15-28):*** *corpus luteum makes progesterone; endometrium becomes glandular and secretes glycogen; if no fertilization the corpus luteum degenerates, hormones fall and menstruation begins on day 28.*

Q7. [SLO 4] Differentiate between human male and female reproductive systems.

Ans:

Male	Female
Testes outside body in scrotum	Ovaries inside the pelvic cavity
Sperms made continuously from puberty (about 400 million / day)	One ovum a month after puberty; oocytes formed in foetus
Accessory glands add semen	No accessory glands for gametes; uterus nourishes embryo
Continuous hormone control with negative feedback	Cyclic hormone control (menstrual cycle)
Fertile throughout life	Cycle stops at menopause (about 50 years)

Q8. [SLO 4] What is infertility? What are the causes of human male and female infertility?

Ans:

- *Infertility: inability to conceive after a year of regular unprotected intercourse.*
- *Male: low sperm count, poor motility, abnormal sperms, blocked ducts, low testosterone, infections, varicocele, high scrotal temperature, smoking, alcohol, stress.*
- *Female: no ovulation (hormonal imbalance, PCOS), blocked oviducts, uterine*

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fibroids or endometriosis, cervical mucus problems, infections, age, stress.

Q9. [SLO 4] Explain in vitro fertilization with diagram.

Ans:

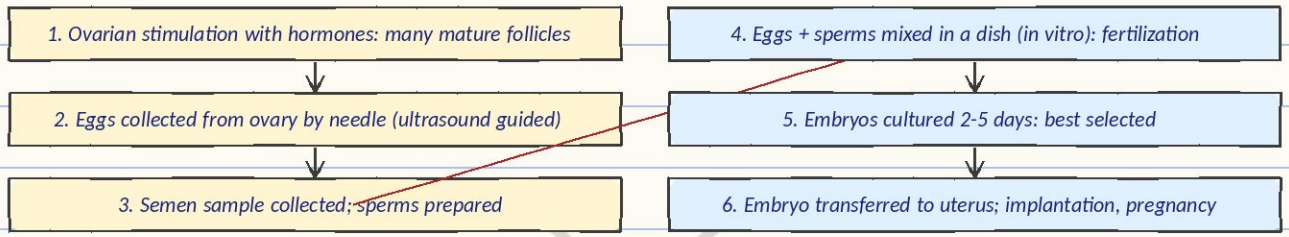


Fig: In vitro fertilization (IVF) - test tube baby technique

- *IVF: fertilization outside the body; hormonal stimulation; egg collection; sperm preparation; mixing in dish; embryo culture; embryo transfer to uterus; pregnancy. The child is a test tube baby.*

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